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B. E. (Eighth Semester) Examination, April-May, 2016

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(New Scheme)

(ET & T Engg. Branch)

POWER ELECTRONICS

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) carry 2 marks. Attempt any two questions (7 marks each) from part (b), (c) and (d) except unit 4. Sketch neat diagram wherever necessary. Sketch waveforms with proper labelling, wherever necessary. Each unit carry 16 marks. Unit-4 having internal choice.

Unit-I

1. (a) Compare Transistors and Thyristors. 2

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(b) Define the following terms :

(i) Holding current

(ii) Latching current

(iii) Break over voltage

(iv) Firing angle 7

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(c) What is the difference between Symmetric and Asymmetric IGBT. Explain the working of punch through IGBT with help of its diagram. 7

(d) Write short notes on following :

(i) Static Induction Thyristor

(ii) Silicon Bilateral Switch 7

Unit-II

2. (a) State the preference of R-C triggering over R triggering. 2

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(b) What are the problems associated with firing of parallel connected SCRs? How can we avoid it? 7

(c) Explain the need of commutation in Thyristor circuits. What are the different methods of commutation? Discuss class F commutation. 7

(d) A relaxation oscillator using an UJT is designed for triggering an SCR.

The UJT has following data :

η = Intrinsic stand off ratio = 0·63

I_P (P-Peaks) = 100 μA, V_p = 15·72

V_1 V_D (Voltage drop) = 0·6 volt

V_V (V-valley) = 1·0 V, I_V = 4 mA,

R_{BB} = Interbase resistance = 5 KΩ,

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Normal leakage current with emitter open = 4 mA

Firing frequency is varied from 2 kHz to 10 kHz.

For $C = 0{\cdot}047\ \mu F$.

Calculate :

(i) Supply voltage V_{BB} (2) R_{max} and R_{min}

(ii) T_{max} and T_{min} (4) R_2 and R_1 7

Unit-III

3. (a) State difference among semiconverter, full converter and Dual converter. 2

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(b) With help of neat sketch and waveform three phase fully controlled Bridge converter with resistive load. 7

(c) What is the difference between symmetric and asymmetric bridge single phase semi converter. Explain it with help of waveforms. 7

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(d) What is the difference between circulating current mode and non circulating current mode dual converter. Explain single phase circulating current mode Dual converter. 7

Unit-IV

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4. (a) Define Duty cycle of chopper. 2

(b) With help of neat ckt diagram and waveform, explain briefly the operation of three phase bridge inverter with resistive load using 180° conduction mode. 10

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Or

With help of neat sketch and waveforms explain the operation of Jones Chopper.

(c) Write short notes on : 4

Chopper control techniques

Unit-V

5. (a) What is the difference between step up and step down cycloconverter? 2

(b) With help of neat sketch and waveform explain single phase full wave A. C. voltage controller. 7

(c) Explain single phase to single phase step up mid point cycloconverter. 7

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(d) Write short notes on TRIAC based AC voltage controller. 7

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